By Invitation

When AI's performance outpaces understanding and safety

Artificial intelligence's impact on mathematics highlights wider risks, says Terence Tao, professor of mathematics at UCLA



IN OCTOBER, OPENAI released more than 700 AI-generated papers solving major open problems in mathematics, many of which are so technical and densely written that no human claims to understand them. To verify such solutions, we use specialised software tools, such as Lean, which can certify that a complex mathematical argument is valid. But in August, an AI model found a security flaw in Lean, which could be exploited to certify that any mathematical statement was proven, whether or not it really was. The flaw was swiftly patched. But it shows that we are in a new reality: even the most trusted verification tools might be gamed in future by a sufficiently capable AI.

In mathematics, this has limited real-world impact, as many of the puzzles we study do not have immediate practical implications. But if this situation is replicated in engineering or in the sciences, we could lose expert human understanding of complex real-world systems, and the mathematics used to model them, just as we are turning over the operation of such systems to autonomous, and potentially misaligned, AI agents. This is a dangerous combination of risks.

The world runs on complex mathematical models. Computational fluid-dynamics models forecast the weather and climate; power grid models balance supply and demand; actuarial models set insurance rates. But models can fail to track reality accurately, due to omissions or errors in the data fed into them, or limitations in their assumptions. These risks are magnified when AI agents are incentivised to push models beyond safe usage parameters.

The advent of powerful AI models, and their incipient deployment in critical real-world infrastructure, magnifies these risks still further. Modern AI is infamous for "reward hacking"—finding loopholes or shortcuts to achieve a given goal—as well as other types of "misalignment", when the goals it pursues do not fully match the intention of the human prompting it. The safety protocols we have traditionally relied upon to protect against catastrophe could meet their match in a powerful AI focused on using whatever means it can to fulfil its objectives. Even such gold standards as a phase III clinical trial might not be immune to AI-powered reward hacking.

The role and value of human understanding is more important than ever

The emerging field of AI safety can provide ways to reduce or mitigate potential harms: by isolating AI agents in "sandboxes", say, or requiring independent audits of their outputs. But one of the strongest defenses against misalignment is robust supervision of AI agents by human experts with a deep understanding of the agent's strengths and weaknesses, the mathematical models it is using, or the domain in which the task resides—and who look for signs of misalignment before approving any outputs.

It is therefore highly concerning that, when faced with the trade-off between speed and performance on one hand, and safety and human understanding on the other, many individuals and firms now promoting the use of AI in real-world applications and in the sciences seem willing to sacrifice the latter in the hopes of making further gains on the former. As recently as 2025, AI labs often collaborated with external academic experts to explore how their tools could augment human-directed research. But now many of the results generated are proudly touted as being nearly fully autonomous, with no human expert involved in supervising the AI or in reviewing its outputs. Indeed, there is often no researcher available from these labs who can present and explain the results, even to other experts in the field.

Fortunately, AI can be used to bolster safety and understanding, rather than undermine it. Experts can use AI tools to stress-test models for vulnerabilities, or perform analyses of competing models. AI-assisted decision-making can be performed in collaboration between human experts and AI agents, with the human understanding of the conclusions reached keeping pace with the speed of output.

Such a human-centred approach offers the best of both worlds: a responsible path to realising the remarkable potential of these tools, while keeping them aligned with genuinely beneficial societal goals. As we enter an era where AI misalignment could affect the internet and the physical world in increasingly negative ways, the role and value of human understanding—in academia and education, in AI labs, in regulation and in the deployment of AI tools—is more important than ever. And the evaluation and mitigation of misalignment risks urgently needs to become a larger part of the AI conversation.